

VPA IB DP CS — Year 1 Semester 2 Overview

VPA IB DP Computer Science — Year 1 Semester 2 Overview

Topics: 3 (Networks) + 4 (Computational Thinking) | Level: SL + HL

Semester Overview

Semester 2 builds on the foundations from Semester 1. Topic 3 (Networks) explores how computers communicate, while Topic 4 (Computational Thinking) develops programming and algorithmic skills essential for the IA and Paper 2. This semester includes intensive IA planning and early development.

Topics Covered

Weeks	Topic	IB Syllabus	Level
1 – 9	Networks	Topic 3 (3.1 – 3.3)	SL + HL
10 – 18	Computational Thinking	Topic 4 (4.1 – 4.4)	SL + HL

Learning Objectives

- Explain how data travels across networks using protocols and packet switching
- Compare network topologies, structures, and security mechanisms
- Construct algorithms using pseudocode and trace tables
- Analyse the efficiency of sorting and searching algorithms
- Apply software development methodology to the IA project

IA Timeline This Semester

Week	IA Milestone	Criterion
1	Client meeting #1 — problem confirmation	A: Planning
4	Success criteria finalised	A: Planning
6	Record of tasks initiated	A: Planning
8	Design document (system architecture, algorithms, UI mockups)	B: Design
12	Core functionality coded and tested	C: Development
16	Development complete; testing begins	C: Development / D: Testing
18	Video evidence recorded; draft Criteria C+D submitted	C + D

Assessment Schedule

Assessment	Week	Format
Topic 3 Progress Check	5	Network diagramming + short answer (30 marks)
Topic 3 End-of-Topic Test	9	Paper 1 style (50 marks)
Programming Progress Check	13	Pseudocode + trace table (25 marks)
Topic 4 End-of-Topic Test	17	Paper 2 style (60 marks SL / 65 marks HL)
Semester 2 Examination	18	Paper 1 + Paper 2 combined (2 hours)
IA Design Document	8	Criterion B (internally marked)

TOK Connections This Semester

Topic 3: The internet is often called a "network of networks"—but what does it mean to say information "travels"? Is data a thing that moves, or is this merely a useful metaphor? How do encryption and trust relate to knowledge claims in digital spaces?

Topic 4: An algorithm is a set of instructions that, followed mechanically, produces a result. In what sense does an algorithm "know" what it is doing? Can a correct algorithm be said to contain knowledge, or is knowledge only in the mind of the programmer?

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